Introduction to Nonlinear Dynamics

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Exam 1 - Solutions

Question Q1. (10 points) **Bifurcations.**

1. (2 points) Let $p \in \mathbb{R}$. Consider the ordinary differential equation

$x_0 = 0$
$x_{1,2} = 1 \pm \sqrt{1-p}$

$$x' = f(x,p) = px + 2x^2 - x^3 - x(x^2-2-p) \tag{1}$$

   on $\mathbb{R}$ and the dynamical system induced by it. Find its steady states.

2. (3 points) Describe the linear stability of the steady states according to the parameter $p \in \mathbb{R}$.

3. (2 points) Sketch the bifurcation diagram of (1).

4. (3 points) Prove that there exists a generic fold bifurcation at a point (x_*, p_*) for (1).

$\hookrightarrow f(x_*,p_*) = 0, \ \partial_x f|_{x_*,p_*} = 0$
$\partial_{xx} f|_{x_*,p_*} \neq 0 \neq \partial_p f|_{x_*,p_*}$

Solution:

1. The steady states of the system satisfy $f(x,p) = 0$. Factoring out x shows that $x_0 = 0$ is an equilibrium for any $p \in \mathbb{R}$. The other steady states satisfy $p + 2x - x^2 = 0$ and are then $x_\pm = 1 \pm (1+p)^{\frac{1}{2}}$. These exist then for $-1 \leq p$.

2. In order to obtain the linear stability of the steady states, we employ the Jacobian of f.

   For x_0,

$$\mathrm{D}f(x_0,p) = p$$

$2x - 3x^2 + p|_{x^*=0} = p$

   implies that x_0 is linearly stable for $p < 0$, non-hyperbolic for $p = 0$ and unstable for $p > 0$. For $x_\pm$, which exist only for $0 \leq 1 + p$ by item 1,

$$\mathrm{D}f(x_\pm,p) = p + 4x_\pm - 3x_\pm^2 = 2x_\pm(1-x_\pm) = 2(\mp\underbrace{(1+p)}_{\geq 0}{}^{\frac{1}{2}} - \underbrace{(1+p)}_{\geq 0}).$$

   Hence, x_+ is stable for any $-1 < p$. We notice that

$$\mathrm{D}f(x_-,p) < 0$$

   if

$$(1+p)^{\frac{1}{2}} - (1+p) < 0,$$

   or equivalently,

$$1 < (1+p)^{\frac{1}{2}}.$$

   This is equivalent to $p > 0$. It follows that the equilibrium x_- is stable for $0 < p$ and unstable for $-1 < p < 0$. At $p = -1$, the equilibria $x_+ = x_- = 1$ are non-hyperbolic.

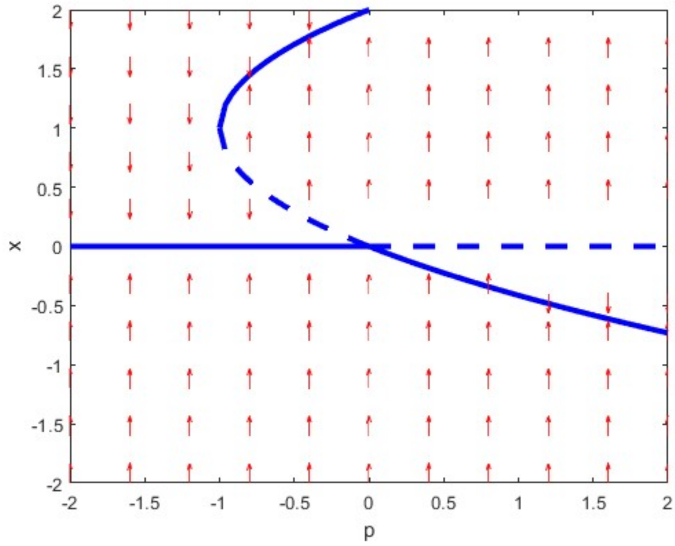

Figure 1: The red arrows indicate the direction of the trajectories. The solid blue lines refer to stable equilibria and the dashed blue line to unstable steady states.

3. The bifurcation diagram is displayed in Figure 1 and obtained from the previous items.

4. The states $x_\pm$ do not exist for $p < -1$ and coincide at $p = -1$. In fact, $p_* = -1$ is a fold bifurcation threshold and $x_* = 1$. We assert it by noticing

$$f(1,-1) = 0, \quad \partial_x f(1,-1) = 0, \quad \partial^2_{xx} f(1,-1) = -2 \neq 0 \quad \text{and} \quad \partial_p f(1,-1) = 1 \neq 0.$$

Question Q2. (10 points) **Nonlinear Stability.**

Consider the following system of ordinary differential equations

$$\begin{aligned} x' &= -y + xz + x^2 y\,, \\ y' &= \quad x + yz - x^3\,, \\ z' &= -z - (x^2 + y^2) + z^3\,. \end{aligned}$$

$$\nabla f = \begin{pmatrix} z+2xy & x^2-1 & x \\ 1-3x^2 & z & y \\ -2x & -2y & -1+3z^2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \to \begin{array}{l} \lambda_1 = -1 \\ \lambda_{2,3} = \pm i \end{array} \quad \begin{array}{l} Re(\lambda_{2,3}) \\ = 0 \end{array}$$

1. (2 points) Prove that the origin is a non-hyperbolic equilibrium.
   $x' = y' = z' = 0$ @ $(0,0,0)$
2. (4 points) Find a quadratic approximation in the form $h(x, y) = ax^2 + bxy + cy^2$ of the center manifold of the equilibrium point at the origin.
   say $z = h(x,y)$ (ansatz) since CM is 2-dim.
   $\dot{z} = h_x \cdot \dot{x} + h_y \cdot \dot{y} = 0$ solve for linearization and a, b, c.
   $z = -x^2 - y^2$
3. (4 points) Determine the local stability of the equilibrium.
   Hint: Use a Lyapunov function or polar coordinates.

Solution:

1. The origin is a steady state of the system. We define

$$f(\bar{x}) = \begin{pmatrix} f_1(\bar{x}) \\ f_2(\bar{x}) \\ f_3(\bar{x}) \end{pmatrix} = \begin{pmatrix} -y + xz + x^2 y \\ x + yz - x^3 \\ -z - (x^2 + y^2) + z^3 \end{pmatrix}$$

   for any $\bar{x} = (x_1, x_2, x_3)$. The linear stability can be studied through the spectral properties of the Jacobian of f on the origin:

$$\mathrm{D}f(0,0,0) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Its eigenvalues are -1 and $\pm i$ for i the imaginary unit. The origin is then a non-hyperbolic equilibrium.

2. From the previous item, we have proven that the center manifold is two-dimensional and the stable manifold is one-dimensional. We consider the quadratic ansatz $z = h(x, y) = ax^2 + bxy + cy^2$. Through the time derivative, we obtain that

$$\begin{aligned} & -h(x,y) - (x^2+y^2) + h(x,y)^3 \\ =& (2ax+by)(-y+xh(x,y)+x^2y) + (bx+2cy)(x+yh(x,y)-x^3) \end{aligned}$$

Using the notation $\mathcal{O}(3) = \mathcal{O}(||(x,y)||^3)$, this is equivalent to

$$(a+1+b)x^2 + (b-2a+2c)xy + (c+1-b)y^2 = \mathcal{O}(3).$$

As such, the coefficients satisfy

$$\begin{aligned} a+1+b &= 0 \\ b-2a+2c &= 0 \\ c+1-b &= 0, \end{aligned}$$

i.e., $a = c = -1$ and $b = 0$. The quadratic approximation of the center manifold is then

$$\left\{(x,y) \,\middle|\, z = -x^2 - y^2, \text{ for } z \in \mathbb{R}\right\}.$$

3. In the previous item, we have proven that near the origin, the trajectories on the center manifold satisfy

$$\begin{aligned} x' &= -y - x(x^2+y^2) + x^2y + \mathcal{O}(4)\,, \\ y' &= x - y(x^2+y^2) - x^3 + \mathcal{O}(4)\,. \end{aligned}$$

We define the Lyapunov function on the plane (x, y),

$$V(x,y) = \frac{1}{2}(x^2+y^2).$$

The Lie derivative of V along the trajectory is

$$\begin{aligned} V'(x,y) &= xx' + yy' \\ &= -xy - x^2(x^2+y^2) + x^3y + xy - y^2(x^2+y^2) - x^3y + \mathcal{O}(5) \\ &= -(x^2+y^2)^2 + \mathcal{O}(5). \end{aligned}$$

Since the z-component is linearly stable, the origin is stable. Alternatively, the proof can be achieved through the use of polar coordinates.

Question Q3. (10 points) **Topics in Dynamical Systems.**

1. (2 points) Let $x' = f(x)$ for $f : \mathbb{R}^n \to \mathbb{R}^n$ and let $y = h(x)$ for a diffeomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$. Determine g such that $y' = g(y)$.

2. (2 points) Define the notion of topological conjugacy between two discrete-time dynamical systems in $\mathbb{R}^n$.

3. (2 points) Assume that $g : \mathbb{T}^2 \to \mathbb{T}^2$ is an ergodic map on the torus $\mathbb{T}^2 = [0, 1)^2$ with respect to the Lebesgue measure. Consider $f(x, y) = x + y$ and let $(x_0, y_0) \in \mathbb{T}^2$ be given. Compute

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f\left(g^j(x_0, y_0)\right).$$

4. (4 points) State the Andronov–Leontovich Theorem.

Solution:

1. The ODE

$$y' = \mathrm{D}h(x)x' = \mathrm{D}h(x)f(x) = \mathrm{D}h(h^{-1}(y))f(h^{-1}(y)) =: g(y)$$

is solved by y and x.

2. A dynamical system $(\mathbb{R}^n, \mathcal{T}, \phi_t)$ is called topologically conjugate to a system $(\mathbb{R}^n, \mathcal{T}, \psi_t)$ if there exists a homeomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$ mapping orbits from the first system to the second system preserving the direction of time and the time parametrization. This is equivalent to

$$\phi_t = h^{-1} \circ \psi_t \circ h$$

for all $t \in \mathcal{T}$.

3. We apply Birkhoff's Ergodic Theorem and obtain

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f\left(g^n(x_0, y_0)\right) = \int_0^1 \int_0^1 x + y \,\mathrm{d}x\mathrm{d}y = 1.$$

4. Consider the ODE

$$x' = f(x, p)$$

for $x \in \mathbb{R}^2$, $p \in \mathbb{R}$, $f : \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}^2$ smooth such that $f(0, p) = 0$ and $x_* = 0$ is a hyperbolic saddle point. Let $\lambda_{1,2}(p)$ be the eigenvalues of x_* with

$$\lambda_1(0) < 0 < \lambda_2(0).$$

Let $\gamma_0(t)$ be a homoclinic orbit to x_* for $p = 0$. Assume the following genericity conditions hold:

(A1) ("non-resonance") We have

$$\sigma(0) := \lambda_1(0) + \lambda_2(0) \neq 0.$$

(A2) ("breaking") The split function

$$\xi(p) = q$$

satisfies $\xi'(0) \neq 0$.

Then a unique limit cycle Γ_q, lying in a neighbourhood of $\gamma_0 \cup \{0\}$, bifurcates from γ_0 upon variation of p. Furthermore, the cycle is stable and exists for $\sigma(0) < 0$, while the cycle is unstable for $\sigma(0) > 0$.

Question Q4. (10 points) **Invariant Measure.**

Let $X = \{e_1, \dots, e_n\}$ be the set of unit vectors in $\mathbb{R}^n$, let $A \in \mathbb{R}^{n\times n}$ be a permutation matrix (i.e. a matrix for which in every row and every column exactly one entry is equal to 1 and all others are equal to 0) and consider the map

$$g : X \to X; \quad x \mapsto Ax. \tag{2}$$

1. (2 points) Show that A maps unit vectors to unit vectors and that the triple $(X, \mathbb{Z}, (g^k)_{k\in\mathbb{Z}})$ is an invertible, discrete-time dynamical system.

2. (2 points) Assume $n = 2$ and $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Let $\mathcal{F}$ denote the power-set of X. Find a probability measure μ on the measurable space $(X, \mathcal{F})$ which is invariant under g.

3. (2 points) Now let $n \in \mathbb{N}$ be arbitrary and let $A \in \mathbb{R}^{n \times n}$ be an arbitrary permutation matrix. Find a probability measure μ on $(X, \mathcal{F})$ which is invariant under g.

4. (2 points) Let μ and ν be g-invariant probability measures on $(X, \mathcal{F})$. Show that

$$\pi_t : \mathcal{F} \to [0, \infty]; \quad B \mapsto \pi_t(B) := t\mu(B) + (1-t)\nu(B) \tag{3}$$

is also a g-invariant probability measure on $(X, \mathcal{F})$ for every $t \in [0, 1]$.

5. (2 points) Assume in addition that $A \in \mathbb{R}^{n \times n}$ is a block-diagonal matrix with at least 2 blocks. Find an infinite family of g-invariant probability measures.

Solution:

1. By definition, the columns of A are unit vectors. Hence the image of a unit vector is a unit vector.

   The triple is now a discrete time dynamical system since it is induced by the iterate of a map, as in Example 1.3 of the lecture notes. Checking the conditions of Definition 1.1 is also fine.

   The system is invertible since A is. Indeed, assume A was not the case, then the columns are linearly dependent. But since they are unit vectors, two columns are equal. This contradicts the fact that every column is also a unit vector by definition of permutation matrix.

2. The uniform distribution $\mu = \frac{1}{2}\delta_{e_1} + \frac{1}{2}\delta_{e_2}$ on X is an invariant measure. This can be seen by checking the condition $\mu(g^{-1}(B)) = \mu(B)$ for every $B \in \mathcal{F}$ (which contains just 4 elements). We have

$$1 = \mu(g^{-1}(X)) = \mu(X) = 1, \tag{4}$$
$$1/2 = \mu(g^{-1}(\{e_1\})) = \mu(\{e_2\}) = 1/2, \tag{5}$$
$$1/2 = \mu(g^{-1}(\{e_2\})) = \mu(\{e_1\}) = 1/2, \tag{6}$$
$$0 = \mu(g^{-1}(\emptyset)) = \mu(\emptyset) = 0. \tag{7}$$

   Alternatively, it can be deduced from the next item.

3. The uniform distribution on $(X, \mathcal{F})$ is invariant: Let $B \in \mathcal{F}$ be arbitrary. Then by definition of the uniform distribution on a finite set, $\mu(B) = \frac{|B|}{|X|}$, where $|\cdot|$ denotes the cardinality of a set. Since A is invertible

$$\mu(B) = \frac{|B|}{|X|} = \frac{|g^{-1}(B)|}{|X|} = \mu(g^{-1}(B)). \tag{8}$$

4. Let $B \in \mathcal{F}$ and $t \in [0, 1]$ be arbitrary. Then

$$\pi_t(g^{-1}(B)) = t\mu(g^{-1}(B)) + (1-t)\nu(g^{-1}(B)) = t\mu(B) + (1-t)\nu(B) = \pi_t(B). \tag{9}$$

   Hence π_t is also g-invariant.

5. Since A is block-diagonal, there exists a $1 \leq k_0 \leq n$ such that $A\{e_1, \ldots, e_{k_0}\} \subseteq \{e_1, \ldots, e_{k_0}\}$ and $A\{e_{k_0+1}, \ldots, e_n\} \subseteq \{e_{k_0+1}, \ldots, e_n\}$. Hence, by the same argument as in item 3 before, the uniform distribution on $\{e_1, \ldots, e_{k_0}\}$ and the uniform distribution on $\{e_{k_0+1}, \ldots, e_n\}$. Hence, by item 4 there is an infinite family of g-invariant probability measures parametrized by $[0, 1]$, which is given by the family of convex combinations of the two.